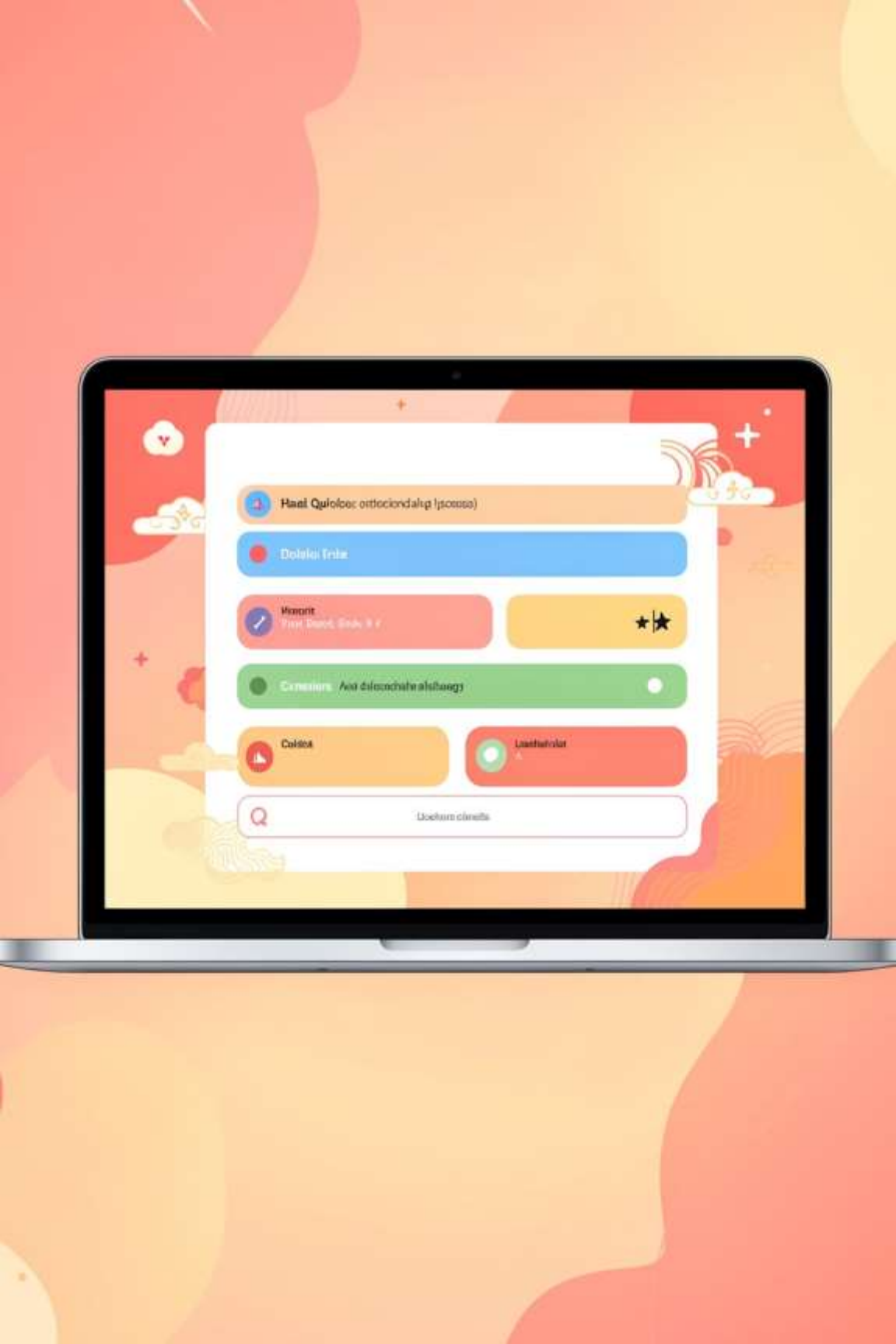




Online Quiz Application

Team Members:

T Devi Surekha

P Sivani Sahitya

College: Pragati Engineering College

Technology Stack: HTML, CSS, JavaScript

Project Overview

Web-Based Quiz System

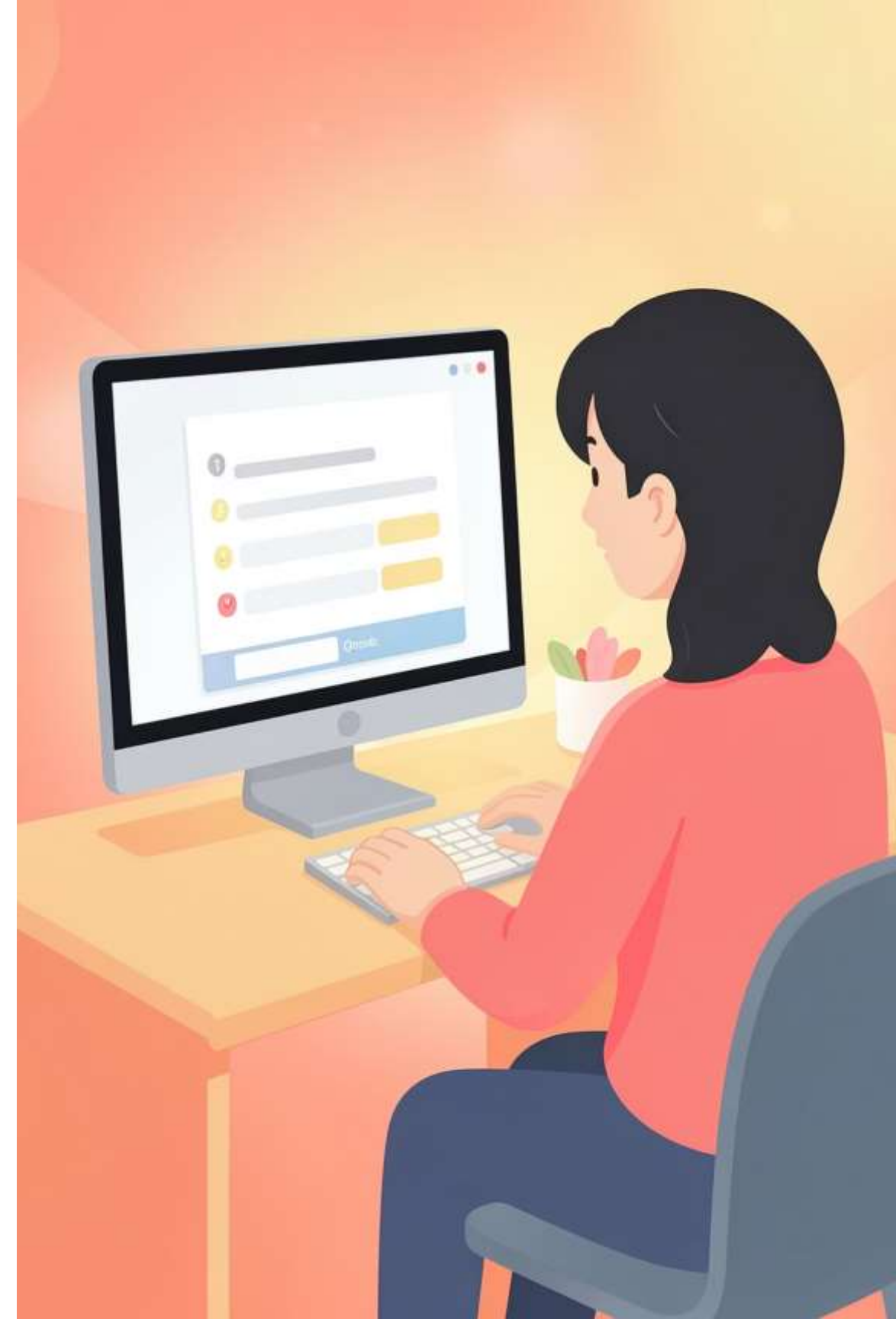
Interactive learning platform built entirely with front-end technologies

Timer-Based Questions

Time-limited questions create urgency and test quick thinking

Real-Time Scoring

Instant feedback shows correct answers and tracks progress



Project Objectives

01

Build Dynamic Quiz Application

Create an engaging, responsive web-based quiz system with smooth transitions and intuitive navigation

03

Improve UI/UX Design

Design clean, modern interface with accessibility features and responsive layout

02

Implement Random Question Selection

Develop algorithm to shuffle questions and answers for varied user experience

04

Provide Instant Feedback

Display correct answers immediately and calculate scores in real-time

Technologies Used



HTML

Structured document foundation with semantic elements for accessibility and SEO



CSS

Styling and animations for responsive design, dark mode toggle, and interactive effects



JavaScript

Core logic for question randomization, timer functionality, and dynamic content updates



LocalStorage

Persistent high score tracking using browser's local storage API

Key Features

Multiple Choice Questions

Four answer options per question with clear visual indicators for selection

Timer-Based Quiz

Fixed time limit per question creates urgency and tests quick thinking

Auto Skip on Timeout

Automatically advances to next question when timer expires

Exit Button

Allows users to quit quiz early and view results

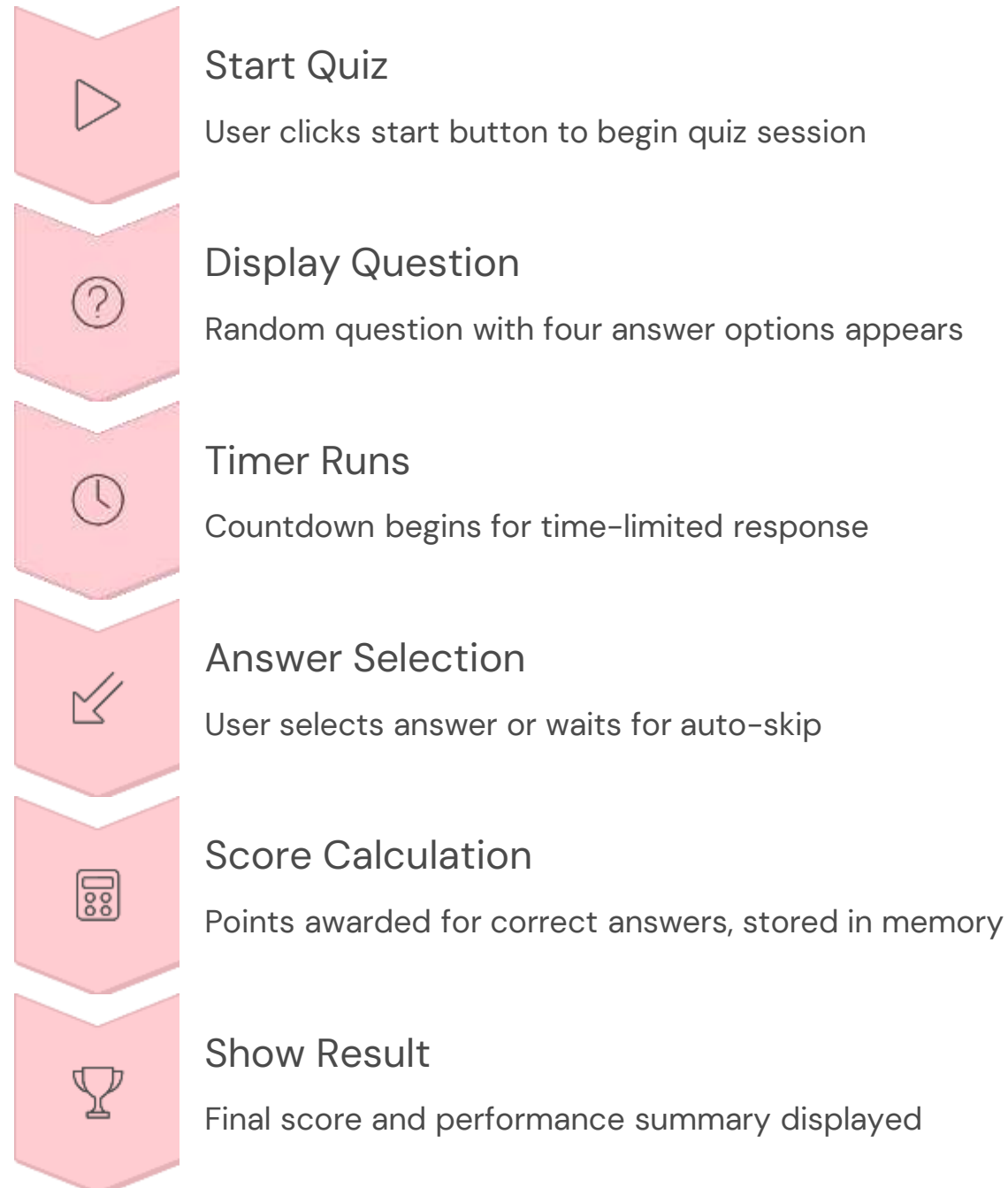
Dark Mode Toggle

Switch between light and dark themes for comfortable viewing

Progress Bar

Visual indicator shows quiz completion percentage

Working Flow



Advantages & Limitations

Advantages

- **User-Friendly Interface:** Intuitive design requires no training
- **Fast Performance:** All logic runs client-side with minimal loading time
- **No Backend Required:** Fully functional without server infrastructure
- **Lightweight:** Small file size ensures quick downloads
- **Offline Capable:** Works without internet connection after initial load

Limitations

- **No Database:** Questions stored in JavaScript arrays, limiting scalability
- **No Login System:** Cannot track individual user progress or history
- **No Data Persistence:** High scores reset if localStorage is cleared
- **Limited Question Types:** Only multiple choice supported
- **Client-Side Only:** Cannot validate answers securely

Future Enhancements



Backend Integration

Connect to Node.js or Python backend for dynamic question loading and score management



Leaderboard System

Track and display top scores across all users with ranking positions



User Authentication

Implement login system with JWT tokens to track individual progress



Mobile Application

Convert to React Native or Flutter for iOS and Android platforms

